

Oliver Fogelin

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Triple citizen (US, UK, Australia) with full work rights in all three countries

EDUCATION

University of Cambridge

Cambridge, UK

BA Computer Science (Hons)

Sep. 2023 – June 2026

- Dissertation: **gvecdb**, a hybrid graph/vector database in OCaml implementing MVCC concurrency control, HNSW indexing from scratch, and AVX-512 SIMD vectorized distance kernels via the OxCaml compiler
- Took modules in **Multicore Semantics and Programming** (lock-free data structures, relaxed memory models, concurrent algorithms), Concurrent and Distributed Systems, Optimising Compilers

EXPERIENCE

Software Engineer Intern

June 2025 – Sep. 2025

Jump Trading

London

- Built firm-wide ClickHouse schema exploration tooling with a web IDE for SQL execution, real-time visualization, and LLM-powered dashboarding. Immediate adoption across dev and trading teams
- Shipped internal Python, Go, and Typescript libraries used across the firm. Received return offer

Y Combinator Summer Fellows Grant

June 2025 – Aug. 2025

Y Combinator

- Selected as 1 of 29 undergrads internationally from over 10,000 applicants (0.29% acceptance) for a \$20k research grant to build open-source developer tools
- Built **stardust**: an embeddable hybrid graph-vector database in C++ on LMDB with Cap'n Proto zero-copy serialization, supporting concurrent graph traversal with HNSW-based approximate nearest-neighbor search

Lead Organiser & Platform Engineer

Oct. 2025 – May 2026

Cambridge Battlecode (Jump Trading title sponsor)

Cambridge, UK

- Built full competition platform from scratch: Rust game engine (PyO3 bindings), nsjail-sandboxed match execution on EC2 Graviton3 (CPU-pinned, AES-256 encrypted bot source), match scheduler dispatching to AWS Batch fleet. 1,500 players from 50+ countries, 5M+ games executed
- Designed protobuf replay format, Phaser 3 visualiser, CLI tool distributed via PyPI (maturin-compiled Rust wheels), tournament bracket system. Co-hosted weekly streams and hosted live finals at Cambridge Union for 60 international finalists

Software Engineer

Jan. 2025 – June 2025

Iterate

London

- Founding team at a company backed by OpenAI, Garry Tan, Index Ventures, and Naval Ravikant, building AI agent products for early-stage startups. Founded by ex-Monzo CTO/Co-Founder
- Architected and shipped full-stack product with Typescript/React, Cloudflare Workers for Platforms, D1, and multi-tenant sandboxed execution

Software Engineer Intern

Aug. 2024 – Oct. 2024

ReactWise (YC S24)

San Francisco

- At 19, sole engineer during the YC S24 batch. Owned entire product (Python/Typescript backend, React frontend, infrastructure, CI/CD) from day zero to production deployment at Pfizer in 10 weeks
- Built full-stack app for Bayesian optimisation of chemical reactions, integrating with pharma high-throughput experimentation workflows and optimising transfer learning in deployment

PROJECTS

YENDOR | *C, Bison, Emscripten/WASM, OpenGL* | [github](#) | [play](#)

Dec. 2025

- **Won 1st place overall** in the Langjam Gamejam (34 entries, run by Austin Henley, CMU professor)
- Designed **nh**, a novel programming language compiled via Bison to C, then to WASM via Emscripten. Built a custom garbage-collected C runtime with OpenGL bindings. Entire game written in nh, all in 7 days

tripos.pro | *Typescript, Python, RAG* | [live](#)

Nov. 2024

- Past paper tracker with RAG-powered semantic search over Cambridge CS exams. Full-stack app used by over half the current Computer Science cohort (~150 students)